

HSC-00 Rev13 - Comprehensive Engineering Baseline (Full)

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HSC-00 REVISION 13

COMPREHENSIVE ENGINEERING BASELINE — FULL DOCUMENT

Founder: Amirbahador Atai

26,000 kg | Main Rotor 26.0 m | Tail Rotor 4.0 m | Capsule 7,000 kg

Revision 13 supersedes Rev12 in full. It retains every section of the Rev12 baseline and the appended licensing package, but corrects the architectural weaknesses identified in technical review — most importantly, it removes main-rotor separation from the core safety case, replaces the unsupported single-figure jettison timeline with a staged and benchmarked one, redesigns the recovery-canopy architecture, adds bottom-up rationale to the capsule mass budget, and adds two sections that were previously missing entirely: an honest Risk Register and a Prior Art Posture statement. All other narrative, mass accounting, interface definitions, requirement sets, and verification-gate structure are retained from Rev12, updated only where the corrected architecture requires it.

DOCUMENT CONTROL

Field	Value
Document Title	HSC Generation 1 — Comprehensive Engineering Baseline
Document Number	HSC-00
Document Type	Master Engineering Specification — Conceptual / IP Baseline
Document Class	Conceptual Engineering / Intellectual Property Development Record
Revision	Rev 13 — Generation 1, 26-ton-class baseline, corrected architecture
Status	[PROPOSED] Preliminary Controlled Engineering Baseline
Development Stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Language	English
Confidentiality	Proprietary Concept Development Material
Founder	Amirbahador Atai

Engineering integrity statement: This document defines a controlled conceptual baseline. Numerical values marked TARGET, ASSUMPTION, STUDY, CALC or OPEN are not demonstrated performance, certification results, or flight-proven values. Critical claims require analytical closure, simulation, ground testing, integrated testing and, where applicable, flight-test evidence.

Revision intent (Rev13): Rev12 established a 26,000 kg / 26.0 m main rotor / 4.0 m tail rotor / 1,000 km range baseline with a survival-capsule and rapid-separation architecture. Rev13 preserves that numerical baseline and the survival-capsule concept in full, but corrects three architectural weaknesses that a technical reviewer or acquiring company’s diligence team would flag immediately: (1) dependence of the safety case on unproven main-rotor separation, (2) an unsupported single-figure separation timeline, and (3) an under-justified 8-canopy recovery concept. It also adds the risk disclosure and prior-art awareness that Rev12 lacked.

TOP-LEVEL TECHNICAL BASELINE (REV13)

Parameter	Rev13 Value	Status
Design mass	26,000 kg	[TARGET]
Main rotor diameter	26.0 m	[TARGET]
Main rotor disk area	530.93 m ²	[CALC]
Disk loading	48.97 kg/m ²	[CALC] — see benchmark table §4.1
Tail rotor diameter	4.0 m	[TARGET]
Tail rotor disk area	12.57 m ²	[CALC]
Capsule mass	7,000 kg	[TARGET] — see bottom-up rationale §8
Range target	1,000 km	[TARGET]
Ideal hover power (study)	≈ 3.57 MW	[CALC]
Shaft power @ FM 0.75	≈ 4.76 MW	[CALC]
Occupancy	10–12 persons	[TARGET / STUDY]
Propulsion	Two turboshaft engines	[REQ / STUDY]
Main rotor separation	Not part of core safety case	[DEFERRED — RESEARCH TRACK, see §12.4]
Recovery canopies	3 primary + 2 reserve (5 total)	[STUDY] — see §14
Command-to-clear timeline	Staged per subsystem	[TARGET/OPEN per stage, see §13]

MASS ACCOUNTING — 26,000 kg (unchanged from Rev12, retained in full)

Mass Group	Planning Mass	Status
Survival capsule, complete	7,000 kg	TARGET
Main airframe excluding capsule	5,000 kg	ASSUMPTION / STUDY
Main rotor, hub and controls	2,100 kg	ASSUMPTION / STUDY
Tail rotor, gearbox and drive	480 kg	ASSUMPTION / STUDY
Engines, transmission, gearboxes and mounts	3,100 kg	ASSUMPTION / STUDY
Fuel system + full usable fuel	3,400 kg	ASSUMPTION / STUDY
Rapid separation system	1,000 kg	ASSUMPTION / STUDY
Avionics, electrical, hydraulic, thermal, control	1,600 kg	ASSUMPTION / STUDY
Operating fluids and consumables	320 kg	ASSUMPTION / STUDY
Growth / configuration reserve	2,000 kg	PROGRAM RESERVE
TOTAL	26,000 kg	CALC / CLOSURE

CAPSULE COMPONENT MASS — 7,000 kg (unchanged from Rev12, retained in full)

Capsule Component	Mass
Primary crashworthy cell / internal roll-cage	1,950 kg
Outer capsule shell / fairing / thermal protection	500 kg
Eight fixed landing supports + energy absorbers	850 kg
Crashworthy seats + restraints, 10-12 positions	700 kg
Emergency exits / doors, frames and seals	250 kg
Distributed multi-canopy recovery system	650 kg
Independent emergency power + batteries	250 kg
Life support / oxygen / environmental	200 kg
Avionics / sensing / communications / locator	250 kg
Separation interfaces + release hardware	350 kg
Internal secondary structure / floor / padding	500 kg
Fire / smoke detection and suppression	150 kg
Water landing / flotation / sealing	150 kg
Inspection / access / replaceable protection	100 kg
Capsule configuration growth allowance	150 kg
TOTAL	7,000 kg

1. PURPOSE AND SCOPE

HSC Generation 1 defines the system-level architecture for a 26-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain rather than reliance on a single emergency device.

Rev13 clarifies the scope in one important way relative to Rev12: **the survival chain is designed to work correctly under the assumption that the main rotor is never separated from the aircraft.** This is a deliberate scoping decision (see §12.4) made because no credible trajectory, gyroscopic, or structural analysis yet exists to support main-rotor separation, and building the core safety case on an unproven capability would undermine confidence in the entire architecture.

The baseline covers the aircraft, occupied survival capsule, rapid selective separation system (non-rotor components), safety initiation paths, deterministic safety authorization, stabilization, recovery systems, impact attenuation, post-event survival and the interfaces required to verify the complete chain. The design philosophy is not “separate everything whenever the aircraft starts descending.” Separation is a controlled emergency mode authorized only when defined conditions are satisfied.

The survival architecture is intended to support multiple emergency outcomes: conventional flight recovery where possible, autorotation where applicable, selective hazard/mass reduction, capsule separation, parachute-assisted descent, energy-absorbing landing and post-landing rescue. The architecture must not require the most extreme survival mode for every failure. The baseline also preserves human emergency authority. Automatic systems may initiate or recommend a survival sequence only within defined authorization envelopes; the crew must retain an independent means to command, inhibit or terminate automatic action where technically and physically possible.

This document is an engineering/IP baseline, not a certification approval. Airworthiness, structural substantiation, parachute qualification, release reliability, occupant injury criteria, flight performance and global patent novelty remain open work items.

2. BASELINE STATUS AND CONFIGURATION CONTROL

Status Label	Meaning in this document
[LOCKED CONCEPT]	Architecture intentionally retained as a baseline design principle; implementation remains subject to engineering verification
[REQ]	Top-level requirement proposed for subsequent requirement allocation
[TARGET]	Numerical or functional development objective; not demonstrated
[ASSUMPTION / STUDY]	Working engineering assumption used for preliminary sizing/accounting
[CALC]	Value calculated from stated assumptions and equations
[OPEN]	Requires analysis, simulation, test, supplier data or design decision
[OPEN — CRITICAL]	An open item with material effect on the core safety case; flagged for priority closure (new in Rev13)
[DEFERRED — RESEARCH TRACK]	Explicitly excluded from the licensable core baseline pending independent research (new in Rev13)
[TBD — DEPENDENCY]	Cannot be responsibly fixed until a dependent design activity closes
[PRIOR ART STATUS: NOT YET SEARCHED]	No novelty conclusion is implied

The baseline shall be configuration-controlled. Any change to MTOW, rotor diameter, capsule mass, occupant count, separation sequence, parachute architecture, landing-support geometry or safety-layer independence shall trigger an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

3. TOP-LEVEL TECHNICAL BASELINE — NARRATIVE

Maximum Take-Off / Design Mass is 26,000 kg [TARGET]. Occupancy is 10–12 persons [TARGET/STUDY]. Flight architecture is a conventional single main rotor plus tail rotor [REQ/STUDY]. Main rotor diameter is 26.0 m [TARGET] with disk area 530.93 m² [CALC] and weight-based disk loading 48.97 kg/m² [CALC] — see the benchmark comparison in §4.1, which situates this figure against real heavy-lift rotorcraft rather than leaving it unexplained. Tail rotor diameter is 4.0 m [TARGET] with disk area 12.57 m² [CALC]. Propulsion architecture is two turboshaft-class engines [REQ/STUDY]. Landing gear is not retractable wheeled gear; instead, eight short, thick fixed mechanical supports integrated with the capsule [TARGET/CONCEPT]. Range target is 1,000 km [TARGET]. Hover endurance is to be derived after power/fuel/thermal closure [OPEN].

The survival capsule is the dedicated occupied final landing boundary [REQ], with a preliminary complete mass of 7,000 kg [TARGET] and bottom-up rationale added in §8. Capsule recovery in Rev13 uses a **3-primary/2-reserve distributed canopy architecture** [STUDY] rather than Rev12’s undifferentiated 8-canopy concept — see §14 for the rationale.

Rapid separation covers selective separation of designated high-mass/fire-hazard components **excluding the main rotor** [REQ/CONCEPT]. The separation development objective is expressed as a staged, per-subsystem timeline rather than a single figure [TARGET/OPEN per stage — see §13].

The 26.0 m rotor is intentionally sized for the 26-ton class; the resulting disk loading of approximately 48.97 kg/m² is, per §4.1, a deliberate hover-margin and downwash-severity trade rather than an unexamined byproduct. Preliminary ideal hover power is approximately 3.57 MW at sea-level standard density; applying an illustrative figure-of-merit assumption of 0.75 gives approximately 4.76 MW rotor power — a study calculation only. Installed engine power must be determined from a complete power-required/power-available model including profile power, induced power, transmission losses, control margins, temperature, density altitude, rotor speed, transients and OEI requirements. The 1,000 km range is a design target; it shall not be presented as achieved range until the mission model closes with aircraft drag, rotor performance, engine specific fuel consumption, fuel capacity, reserves, payload, temperature/altitude and operational constraints.

4. AIRCRAFT GEOMETRY AND CONFIGURATION

The geometry below is a preliminary architecture sized around the 26-ton class. Dimensions other than rotor diameters are deliberately kept as ranges or open items until aerodynamic, structural, cabin, capsule and systems layouts are reconciled.

Geometry Item	Preliminary Definition	Status
Overall length	Approx. 16.5-18.0 m	[TARGET/STUDY]
Maximum fuselage width	Approx. 4.0-4.5 m	[TARGET/STUDY]
Overall height	Approx. 4.8-5.3 m	[TARGET/STUDY]
Main rotor diameter	26.0 m	[TARGET]
Tail rotor diameter	4.0 m	[TARGET]
Rotor arrangement	Single main rotor + tail rotor	[REQ/STUDY]
Landing supports	8 fixed short mechanical supports under capsule	[TARGET]
Retractable landing gear	None	[LOCKED CONCEPT]
Capsule position	Central occupied boundary integrated within primary airframe	[LOCKED CONCEPT]
Capsule normal-flight load path	Dedicated primary attachment system with controlled emergency release	[REQ]
Capsule emergency load path	Capsule becomes independent final landing element after verified clearance	[REQ]

The eight landing supports are not simple struts attached to a thin capsule shell. The intended load path is: ground contact → foot/ground interface → progressive energy absorber → lower capsule load path → primary capsule structure → occupant restraint/seat energy management. Structural sizing shall consider vertical, longitudinal, lateral and torsional loads, slope, uneven terrain, soft soil, asymmetric touchdown and failed-leg cases.

4.1 Rotor Disk Loading — Benchmark Comparison (new in Rev13)

Rev12 presented the 48.97 kg/m² disk loading figure without context, which invites a reviewer to interpret it as either an error or an unexplained anomaly. Rev13 adds the comparison a diligence engineer will make regardless, and states the design rationale explicitly:

Aircraft	MTOW (approx.)	Main Rotor Diameter	Disk Loading (approx.)
Mi-26	56,000 kg	32.0 m	~55 kg/m ²
CH-53K	38,400 kg	24.1 m	~58-60 kg/m ²
HSC Gen1 (Rev13)	26,000 kg	26.0 m	48.97 kg/m²

Design rationale: HSC’s disk loading is deliberately lower than comparable heavy rotorcraft. This is presented as a trade for greater hover power margin — useful given the added mass and complexity of the capsule and separation systems — and for reduced downwash severity near the fixed landing supports during ground operations. This framing turns what would otherwise be a flagged anomaly into a stated, defensible engineering decision, to be confirmed or revised once Document 063 (Main Rotor Disk Loading Performance Study) closes.

5. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

The architecture retains the six-layer functional survival philosophy. The layers are functionally distinct. Duplication alone shall never be used as evidence of independence. Independence shall be assessed for power, signal routing, sensors, processors, physical location, structural vulnerability, fire, EMI/lightning, common-cause failures and human interaction.

Layer	Function	Primary Purpose
L1	Primary automatic emergency detection and initiation	Detect catastrophic states and initiate a validated survival sequence when authorization conditions are met
L2	Independent backup initiation	Provide a second physically/electrically segregated automatic path if L1 is unavailable
L3	Independent manual/crew authority	Allow crew initiation, inhibit and emergency authority without dependence on L1/L2
L4	Rapid selective physical separation	Separate designated non-rotor mass/fire hazards from the capsule using controlled release mechanisms
L5	Direct-impact survival	Protect occupants through capsule structure, restraints, seats and eight energy-absorbing supports
L6	Recovery and post-event survival	Deploy recovery system, stabilize landing, locate occupants and support post-landing survival/rescue

The design principle is: AI or advanced prediction may assist detection, classification and decision support; deterministic safety logic authorizes safety-critical actuation. Confidence or model output shall never be treated as equivalent to safety authorization. This principle underlies the strongest IP claim in the package — see §17, IP-A.

6. DETERMINISTIC SAFETY AUTHORIZATION AND SAFETY KERNEL

The Safety Kernel is the deterministic authorization boundary between emergency detection/decision logic and physical survival actuation. Its purpose is to prevent a computationally complex or uncertain decision layer from directly commanding hazardous separation without satisfying explicit safety preconditions.

The Safety Kernel shall evaluate, as applicable: altitude above terrain and minimum recovery altitude; airspeed and dynamic pressure envelope; pitch, roll, yaw and angular-rate limits; rotor RPM, rotor state and rotor control state; engine/fire/fuel hazard state; structural health and confirmed structural failure state; capsule attachment health and release readiness; separation trajectory clearance; potential collision between released components and capsule/parachutes; parachute deployment envelope and recovery-system availability; landing support availability and degraded configurations; sensor validity and disagreement; power availability for release and recovery systems; communication availability where applicable; manual crew authority and emergency override state; rollback/fallback availability.

If the safety conditions are not satisfied, the system shall inhibit the unsafe action or transition to a defined degraded mode. Safety and physical feasibility take priority over optimization or mission continuation. No software model, AI classifier, optimizer or predictive system shall bypass the Safety Kernel, aircraft protection systems, interlocks, equipment limits or human emergency authority.

7. CONTROLLED RAPID SEPARATION SYSTEM

7.1 Separation philosophy

The rapid separation system is a selective, staged, controlled mechanism for reducing the mass, fire load and rotating/energetic hardware that must remain associated with the occupied capsule during the final descent. It is not a blanket requirement to release every aircraft component in every emergency. The system shall use state-dependent release logic. Components shall be designated as jettisonable only after structural, aerodynamic, dynamic, fire, control, trajectory and collision analysis demonstrates that release is acceptable within a defined envelope.

7.2 Candidate separable components (revised in Rev13)

- Main engines and associated high-mass engine installation interfaces, where practical.
- Selected transmission/drive components where release is structurally and dynamically acceptable.
- Tail rotor, tail-rotor gearbox and drive shafting where release is safe.
- Primary fuel tanks and major fuel lines that constitute a significant fire hazard, after safe fuel isolation.
- Selected external structural panels or high-mass attachments that do not form part of the capsule primary load path.
- Other components only through configuration-controlled hazard and trajectory analysis.

Main rotor blades and hub components are explicitly excluded from the core candidate list in Rev13. See §12.4 for the rationale and for the deferred research-track status of this capability.

7.3 Mandatory release protections

Positive mechanical retention during normal flight; electrical/logic inhibits against inadvertent release; independent command paths; defined arm/fire sequence; release-state sensing; fuel and energy isolation before or coordinated with release; asymmetric/partial-release detection; post-release trajectory monitoring where practical; capsule-clear verification before parachute authorization; defined failed-release and rollback/fallback behavior.

8. OCCUPIED SURVIVAL CAPSULE

The occupied survival capsule is the central safety boundary of HSC Generation 1. During normal flight it is structurally integrated into the aircraft through controlled primary attachments. During a validated catastrophic survival sequence, it is intended to become an independent occupied landing element.

Capsule Parameter	Definition	Status
Occupancy	10–12 persons	[TARGET/STUDY]
Complete capsule mass	7,000 kg planning target	[TARGET]
Primary structure	Continuous CFRP roll-cage with metallic/titanium nodes and controlled joints	[CONCEPT]
Seats	Energy-absorbing seats with five-point/appropriate restraints	[TARGET]
Landing supports	8 short, thick fixed mechanical supports with progressive energy absorption	[TARGET]
Parachute architecture	3 primary + 2 reserve distributed canopies	[STUDY — revised, see §14]
Capsule emergency systems	Independent emergency power, sensing, locator, communications and essential life-support	[TARGET]
Capsule clearance	Verified before recovery deployment	[REQ]
Post-landing stability	Eight-point support architecture with asymmetric/failed-leg analysis	[REQ]

The capsule shall be designed as a real structural system rather than a passenger cabin merely surrounded by a shell. The engineering program shall address load paths, crash loads, fatigue, damage tolerance, joints, attachment fittings, seat loads, floor loads, fire/smoke, environmental exposure, water/ditching provisions where required, emergency egress and occupant injury criteria. The capsule shall retain sufficient independent capability after separation to support recovery and rescue. Essential systems should not depend on the main airframe power bus after separation.

8.1 Capsule mass — bottom-up rationale note (new in Rev13)

The 7,000 kg figure is retained from Rev12 as a planning target pending the bottom-up closure defined in Document 113 of the Master Document Tree, but Rev13 adds the context a buyer or reviewer needs to judge its plausibility rather than leaving the number unexplained:

- Per-occupant mass at 10–12 occupants is approximately 580–700 kg — materially higher than a standard aviation cabin allocation, because the capsule mass includes a fully independent structural cell (not shared load path with the airframe), the eight-point external landing/energy-absorption interface, distributed-parachute hard points, and independent emergency power and life support, none of which a conventional cabin carries.
- The nearest relevant structural analog is a crewed spacecraft return capsule, where fully-equipped per-occupant mass (including independent life support and thermal protection) commonly runs in a comparable or higher range — a closer comparison class than a helicopter cabin.
- This remains an explicit planning allocation, not a verified figure; Document 113 (Capsule 7000 kg Mass Budget Bottom-Up) is still required before this number can be presented as engineering-closed.

9. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

HSC Generation 1 uses eight short, thick, fixed mechanical supports located below the survival capsule. The concept is intentionally different from conventional retractable landing gear: the supports are part of the survival capsule's impact-attenuation system and remain with the occupants after separation.

Eight supports provide load distribution and redundancy against uneven terrain and partial failure. Each support shall include a defined mechanical energy-absorption function rather than relying solely on elastic deformation. Replaceable or inspectable sacrificial/crush elements should be considered to improve post-event maintainability. The attachment system shall transmit controlled loads into the capsule's primary lower structure. The system shall be analyzed for vertical, lateral, longitudinal and combined impact loads. Asymmetric landing with one or more failed/degraded supports

is a required analysis case. Terrain cases shall include hard flat ground, sloped ground, uneven ground and soft/low-bearing-strength surfaces. Ground clearance and rotor-downwash interaction shall be evaluated. Post-landing stability, tipping resistance and rescue access shall be included.

Preliminary capsule impact energy follows $E = 1/2 m v^2$. For a 7,000 kg capsule, illustrative kinetic energies are approximately 126.0 kJ at 6 m/s, 224.0 kJ at 8 m/s and 350.0 kJ at 10 m/s. These are not allowable impact energies; they are sizing illustrations only. Actual survival depends on energy absorption, occupant pulse, attitude/rates, terrain and structural response.

10. EMERGENCY STATE MACHINE

HSC Generation 1 shall use an explicit emergency state machine rather than a single binary “normal/emergency” flag:

1. NORMAL FLIGHT — all safety layers monitored; release hardware mechanically locked.
2. ANOMALY DETECTED — sensor/model detects a potential abnormal condition.
3. CONFIRMATION — independent evidence and validity checks determine whether the condition is real.
4. IMPACT ASSESSMENT — estimate control authority, altitude, airspeed, fire/structural risk, rotor/engine state and available recovery modes.
5. RECOVERY ATTEMPT — use conventional control, engine redundancy or autorotation where safe and appropriate.
6. SURVIVAL AUTHORIZATION — Safety Kernel evaluates whether selective separation/capsule recovery is permitted.
7. STAGED SEPARATION — execute only the designated release set for the specific emergency state (non-rotor components only, per §7.2).
8. CAPSULE CLEARANCE — verify physical separation and identify any attached components.
9. RECOVERY DEPLOYMENT — authorize parachute/recovery system within validated envelope.
10. DESCENT MANAGEMENT — monitor attitude, rates, canopy state and landing-support readiness.
11. LANDING ATTENUATION — deploy/condition energy absorbers as required.
12. POST-LANDING SURVIVAL — stabilize, locate, communicate, protect occupants and support rescue.
13. AUDIT — preserve event data, release states, sensor states and command history for engineering analysis.

Representative state-dependent behavior: an isolated engine failure should not automatically trigger capsule separation; a major fire may prioritize fuel isolation and selective separation; loss of control may require a different separation envelope; structural damage may require immediate transition; a combined catastrophic event may require the full survival chain.

11. INDEPENDENCE, FAILURE MODES AND COMMON-CAUSE CONTROL

The survival architecture introduces new hazards as well as new protections. The following failure classes shall be treated as explicit design cases:

Failure / Hazard	Required Design Response
Unintended separation	Mechanical locks, electrical inhibits, command validation, monitoring and quantitative safety objective
Partial separation	State sensing, asymmetric-load analysis and defined degraded recovery mode
Asymmetric separation	Dynamic analysis, CG/inertia update and post-release control assessment
Release mechanism failure	Independent path or defined safe fallback; no assumption of perfect release
Sensor loss / bad data	Validity monitoring, redundant/diverse sensing and safe-state logic
Automatic detection failure	Independent backup and manual crew initiation
Power-bus failure	Independent survival power where required for critical functions
Communication loss	Local deterministic safety logic; communication shall not be the sole authorization path
AI/model failure	Deterministic safety logic remains authoritative
Fire during separation	Fuel isolation, fire containment and release sequencing
Released component collision	Trajectory analysis, clearance envelopes and controlled sequencing
Parachute deployment outside envelope	Hard inhibits and validated deployment conditions
Landing-support failure	Eight-support redundancy and asymmetric/failed-leg structural cases
Common-cause EMI/lightning/fire/structural damage	Dedicated common-cause analysis and physical/electrical segregation

Independence is not established merely because two boxes, sensors or switches exist. The verification program shall assess common power, common wiring routes, common structure, common software, common sensor physics, common fire zones and common environmental vulnerabilities.

12. CORE SAFETY CASE — REBUILT WITHOUT MAIN-ROTOR SEPARATION (revised in Rev13)

12.1 Why this changed

Rev12 listed the main rotor as a “candidate separable component,” conditional on future analysis. In practice, this created an implicit dependency: several narrative passages in Rev12 implied that in the most catastrophic emergency states, full survival relied on the possibility of rotor separation. A 26 m rotor carries enormous rotational kinetic energy and gyroscopic moment; no credible 6-DOF or structural trajectory analysis yet exists to show that severing and clearing it from a descending capsule and its parachutes can be done safely. Leaving this dependency in the baseline gives any competent technical reviewer grounds to discount the rest of the document’s rigor.

12.2 The corrected principle

The survival chain is designed to work assuming the main rotor is never separated.

12.3 Revised primary survival chain

1. Controlled flight recovery / autorotation (unchanged from Rev12).
2. Selective separation of non-rotor hazards only: engines, fuel tankage (post-isolation), tail rotor/drive assembly, designated fire-load panels.
3. Capsule release from the airframe — the capsule departs from beneath/behind the still-attached (or autorotating) main rotor disk, not through it.
4. Capsule-only descent under the distributed recovery canopy system (§14).
5. Impact attenuation via the eight fixed supports (§9).
6. Post-landing survival and locate/rescue (§16).

12.4 Main-rotor separation — deferred research track

Main-rotor separation is retained in the document as a labeled, walled-off future research item, explicitly excluded from the licensable core baseline and from the primary IP claims (§17). This is a deliberate configuration-control decision: it prevents a buyer’s diligence team from being able to discount the entire package on the strength of its single weakest claim, while preserving the option to reintroduce a rotor-separation capability later if and when independent 6-DOF, structural and trajectory analysis (Document 080, HSC Rotor Emergency Release Feasibility Study) demonstrates it is

achievable.

13. STAGED SEPARATION TIMELINE (replaces Rev12's single ≤ 6 s target)

Rev12 proposed a single " ≤ 6 s from validated command to verified capsule-clear state" target with no supporting rationale. Rev13 replaces this with a staged, per-subsystem timeline that shows its work and flags its own weakest link rather than hiding it:

Stage	Action	Target Window	Confidence Basis
S0	Validate emergency state; arm authorized release path; isolate fuel/electrical/energy hazards	0.0–0.5 s	Comparable to modern ejection-seat arming logic
S1	Fuel isolation and designated fire-hazard separation	0.5–1.8 s	Comparable to certified crash-fuel-shutoff systems in existing rotorcraft
S2	Engine and tail-rotor drive separation (non-main-rotor)	1.8–3.5 s	Comparable to external cargo-release hardware timing, scaled up
S3	Capsule-to-airframe release	3.5–5.0 s	[OPEN — CRITICAL] — no direct engineering analog exists; flagged honestly rather than asserted
S4	Capsule clearance verification	5.0–5.8 s	Sensor-response-limited, not mechanism-limited
S5	Recovery canopy authorization	After S4, gated on verified clearance	Not time-gated; correctness prioritized over speed

If detailed architecture cannot safely achieve these windows, targets shall be revised under configuration control rather than achieved by reducing verification rigor. Partial and asymmetric separation remain first-class emergency states at every stage — the system shall not assume that every release mechanism functions perfectly.

14. RECOVERY ARCHITECTURE — 3 PRIMARY + 2 RESERVE CANOPIES (revised in Rev13)

Rev12 proposed an undifferentiated 8-canopy distributed system with no stated failure-tolerance rationale. Rev13 replaces this with a structure closer to flight-proven capsule-recovery precedent:

Recovery Mode	Use Case	Status
Controlled flight recovery	Recoverable control/propulsion disturbance	[REQ/STUDY]
Autorotation	Engine/power loss when altitude/rotor state permit	[REQ/STUDY]
Selective mass/hazard separation	Fire, structural or combined emergency requiring reduction of carried hazard/mass	[REQ/CONCEPT]
Capsule separation	Catastrophic state where independent occupied recovery is preferable	[REQ/CONCEPT]
3 primary recovery canopies	Primary post-separation vertical recovery	[STUDY — revised from 8 to 3+2]
2 reserve canopies	Deployed only on detected primary-canopy failure, not by default	[STUDY — new in Rev13]
Eight energy-absorbing supports	Final ground impact attenuation	[REQ/TARGET]
Optional airframe recovery parachute	Residual-airframe recovery or unmanned asset protection	[OPTION/TRADE STUDY]
Post-landing survival	Locator, communications, lighting, emergency power and rescue interface	[REQ]

Rationale for the change: three larger primary canopies, sized to the 7,000 kg capsule using standard aerospace descent-rate targets, are directly preceded by crewed spacecraft return-capsule practice (three-main architectures at comparable or greater mass). Fewer, larger canopies reduce inflation-sequencing complexity and inter-canopy collision risk at low altitude — the failure mode most likely in a real emergency, since most rotorcraft accidents occur during low-altitude phases of flight. Two independent reserve canopies provide redundancy without multiplying simultaneous-inflation risk. This directly addresses the weakest point of Rev12's recovery concept.

Explicit open item retained: the minimum deployment altitude at which this (or any) capsule recovery system can be

relied upon is **[OPEN — CRITICAL]**. Rev13 does not claim this is solved; it states plainly that low-altitude emergencies may fall outside any capsule-recovery envelope regardless of canopy count, and that this must be closed analytically (Document 159, Capsule Parachute Sizing Analysis) before any survivability claim is made for low-altitude scenarios.

15. PRINCIPAL SYSTEM INTERFACES

Interface	Definition	Control Requirement
CAP-AF	Capsule ↔ airframe structural attachment	Normal-flight load path + controlled emergency release + positive retention
CAP-PAR	Capsule ↔ parachute system	Deployment only after validated capsule state and clearance
CAP-LG	Capsule ↔ eight landing supports	Primary crash/landing load path with energy absorption
JET-PWR	Independent release power	Segregated from common single-point power failure where required
JET-CMD	Automatic/backup/manual commands	Independent paths and protected actuation
JET-SNS	Release-state sensing	Positive confirmation of attached/released state
FUEL-ISO	Fuel isolation ↔ separation	Isolation before or coordinated with release according to hazard analysis
REC-LOC	Post-landing locator/communications	Independent emergency power and antenna strategy
AV-CTRL	Avionics/control ↔ Safety Kernel	Deterministic authorization boundary

All interfaces crossing survival boundaries shall be documented in Interface Control Documents and assessed for common-cause vulnerability. Connector, harness, hydraulic, fuel and structural interfaces shall be designed so that emergency separation does not create an uncontrolled hazard.

16. VERIFICATION, VALIDATION AND EVIDENCE PROGRAM

The following evidence gates are required before any high-confidence safety or performance claim is made:

Gate	Evidence Objective
G0	Architecture Freeze — confirm safety philosophy, system boundaries, interfaces and emergency states
G1	Analytical Closure — mass, CG, inertia, rotor sizing, power, fuel, structural load paths and separation kinematics
G2	Subsystem Demonstration — release mechanisms, locks/inhibits, energy absorbers, parachute components, sensors and independent power
G3	Integrated Ground Demonstration — end-to-end sequence without flight exposure; verify timing, state sensing and capsule clearance
G4	Dynamic / 6-DOF Simulation — vehicle and capsule dynamics through representative emergency states, partial/asymmetric release and parachute inflation
G5	Structural / Crash Testing — capsule, attachment points, seats, landing supports, energy absorbers and occupant-protection evidence
G6	Recovery System Testing — canopy deployment, inflation loads, failure cases, stability and landing performance
G7	Flight Development — controlled flight tests within approved envelope; validate normal and emergency recovery behavior
G8	Certification / Qualification — evidence package against the selected certification basis and applicable standards

Terms shall be used carefully: “specified” means defined in the design; “analyzed” means a documented analytical result; “simulated” means supported by a model; “tested” means demonstrated in a defined test; “verified” means evidence shows compliance with a requirement; “validated” means suitability has been established for the intended use; “qualified” and “certified” require the applicable formal processes.

Rev13 note: as of this revision, no gate beyond G0 (Architecture Freeze) is complete. This is stated plainly rather than

implied otherwise — see §18, Risk Register.

17. ENGINEERING / IP VALUE SURFACES (narrowed and strengthened in Rev13)

Rev12 listed nine candidate IP elements at roughly equal weight, which dilutes attention and invites a diligence team to discount the whole set once the weakest are found unpatentable or over-broad. Rev13 concentrates on the three most defensible claims and retains the rest as supporting system narrative rather than standalone claims.

17.1 Primary claim candidates

IP-A — Deterministic Safety Kernel gating AI-assisted emergency prediction from physical actuation. The combination of (a) AI/ML-based anomaly prediction, (b) a hard deterministic precondition layer, and (c) selective, state-dependent (not all-or-nothing) actuation is a specific systems-engineering combination, not a single known device. This is assessed as the strongest claim in the package.

IP-B — Eight-point fixed capsule-integrated landing support with independent asymmetric/partial-failure state management. Combining the structural configuration (eight-point, capsule-integrated, non-retractable, energy-absorbing) with explicit failure-state detection and degraded-recovery logic as a single claim is narrower and more defensible than treating the hardware and the logic as separate elements.

IP-C — State-dependent, non-rotor selective separation sequencing for an occupied survival capsule. By explicitly excluding main-rotor separation (§12), this claim is narrowed to something closer to precedented “selective jettison” systems with a specific new element: capsule-centric, staged, state-dependent sequencing logic (§13) rather than an unbounded claim over “separate anything.”

17.2 Supporting system narrative (not standalone patent candidates)

The six-layer survival taxonomy (§5), the multi-mode recovery chain (§14), state-aware partial/asymmetric-release management, the optional residual-airframe recovery concept, and the immutable emergency event record are retained as descriptive system architecture. They strengthen the buyer’s understanding of the overall product but are not presented as independently patentable — a narrower, honestly-scoped claim set is more credible to patent counsel than nine equally-weighted assertions.

18. RISK REGISTER (new in Rev13)

Risk	Severity	Current Status
Capsule-to-airframe release (Stage S3, §13) has no direct engineering precedent	High	Open research item, explicitly flagged
Low-altitude canopy deployment envelope may exclude many real accident profiles	High	Open, needs minimum-altitude closure study (Document 159)
7,000 kg capsule mass not yet bottom-up verified	Medium	Planning allocation only, Document 113 pending
No 6-DOF simulation evidence exists yet for any separation stage	High	Gate G4 not started
No professional prior-art search has been conducted	Medium	See §19
Certification basis (CS-29 analog) not formally engaged with any regulator	Medium	Conceptual reference only
Main-rotor separation capability, if later reintroduced, remains structurally unproven	Medium	Deferred to research track, §12.4

A buyer’s engineering diligence team will identify these risks regardless of whether they are disclosed. Naming them here, in the founder’s own words, is what makes the surrounding technical claims credible rather than promotional.

19. PRIOR ART POSTURE (new in Rev13)

This section does not assert novelty conclusions — that requires a professional prior-art search — but names the nearest known analog classes so a reviewer sees baseline awareness rather than legal naivety:

- **Ejection-seat-equipped rotorcraft.** Existing rotorcraft designs (e.g., certain Soviet/Russian attack helicopters) use rotor-blade-severing explosive charges paired with a crew ejection seat. This is directly relevant prior art for

any main-rotor-related separation claim, which is precisely why Rev13 removes main-rotor separation from the core claim set (§12, §17).

- **Crewed spacecraft return capsules.** Multi-canopy recovery, crashworthy seating and independent post-landing life support in existing crewed spacecraft are relevant prior art for the capsule-recovery and life-support elements of this baseline. Rev13's IP language (§17) is narrowed to the rotorcraft-specific combination rather than the general capsule-recovery concept, which is not novel on its own.
- **Existing crash-attenuating landing gear / energy-absorbing strut systems** in certified rotorcraft are relevant to the eight-support claim (IP-B). Novelty must rest on the capsule-integration and failure-state-management aspects specifically, not on energy-absorbing landing gear in general.

A professional prior-art search against IP-A, IP-B and IP-C specifically (not the whole document) remains a prerequisite before any patent filing or broad data-room disclosure.

20. ENGINEERING BASIS AND EXTERNAL REFERENCE NOTES

The baseline is derived primarily from the founder's prior LSH Document 00 Rev3.2 and the higher-maturity rapid-jettison Generation 1 document. The previous baseline established the six-layer survival philosophy, independent automatic/manual initiation, controlled separation, eight landing supports, the capsule-as-final-landing-boundary concept, and the need to treat partial/asymmetric separation as first-class states. Rev12 updated the Generation 1 planning point to 26,000 kg, 26.0 m main rotor, 4.0 m tail rotor and 1,000 km range target. Rev13 preserves this numerical baseline in full and corrects the architectural dependencies described above.

Current EASA CS-29 material was reviewed as an external engineering reference. The EASA material emphasizes that jettisonable configurations require satisfactory separation characteristics, including clear trajectory, no damaging interaction, no adverse/uncontrollable aircraft reaction, acceptable post-jettison stability/control and acceptable post-jettison performance; the verification program therefore treats these as core analysis/test concerns rather than assuming safe release. External reference: EASA, Easy Access Rules for Large Rotorcraft (CS-29), Amendment 12, March 2026.

21. V13 DESIGN PHILOSOPHY — CONTROLLED SURVIVAL, NOT A SINGLE DEVICE

The defining engineering proposition of HSC Generation 1 remains the integration of multiple survival mechanisms into one state-aware architecture. The aircraft is not intended to depend solely on a parachute, solely on autorotation, solely on a detachable cabin or solely on automatic software. Rev13 strengthens this proposition by removing the one dependency (main-rotor separation) that most directly contradicted it: a safety case that quietly needs its riskiest, least-proven capability to succeed is not actually a layered safety case.

The intended chain is: PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE (NON-ROTOR) → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT → SURVIVE/LOCATE/RESCUE → AUDIT.

The strongest future evidence will come from a closed mass model, rotor/propulsion analysis, 6-DOF simulation, deterministic safety logic, separation mechanism tests, capsule structural tests, parachute tests and progressively integrated demonstrations. Document disposition: Rev13 is suitable as the controlled conceptual/master engineering baseline for the next engineering phase, and is materially more credible to a technical acquirer than Rev12 because it discloses its own weakest points rather than presenting an appearance of completeness. It should not yet be presented as a certified aircraft specification or as proof of achieved 26-ton performance, 1,000 km range or survival probability.

22. DETAILED AIRCRAFT MASS BREAKDOWN (unchanged from Rev12, retained in full)

Main Airframe — 5,000 kg: Fuselage primary load-bearing structure 1,450 kg; frames, bulkheads and longerons 650 kg; tail boom and tail structural assemblies 520 kg; engine and transmission support structure 380 kg; rotor mast/transmission support integration structure 260 kg; external skins, fairings and aerodynamic panels 310 kg; doors, access panels and hinges 190 kg; firewalls, barriers and structural fire protection 170 kg; structural fittings, lugs, fasteners and local reinforcements 260 kg; cabin/service floor structure outside capsule boundary 210 kg.

Main Rotor System — 2,100 kg: Main rotor blades (26.0 m system) 980 kg; rotor hub and blade retention/grips 280 kg; mast and rotor-head structural components 260 kg; swashplate and pitch-control hardware 170 kg; pitch links, control rods and bearings 110 kg; rotor vibration/dynamic-control hardware 80 kg; rotor lubrication/inspection/access hardware 40 kg; configuration allowance 40 kg.

Tail Rotor System — 480 kg: Tail rotor blades (4.0 m system) 145 kg; tail rotor hub and blade retention 65 kg; tail rotor gearbox 95 kg; drive shafting, couplings and bearings 60 kg; tail rotor controls and pitch mechanism 35 kg; tail structure/interface hardware 25 kg.

Engines and Main Transmission System — 3,100 kg: Engine #1 complete installed 700 kg; Engine #2 complete installed 700 kg; engine accessories and engine-control hardware 180 kg; main gearbox/combining transmission 650 kg; intermediate/reduction transmission elements 180 kg; drive shafts and couplings 100 kg; lubrication, oil cooling and filtration 80 kg; engine/gearbox mounts and vibration isolation 60 kg; exhaust and hot-section installation hardware 50 kg.

Fuel System + Full Usable Fuel — 3,400 kg: Usable fuel mass 2,100 kg; fuel tanks and crash/fire-resistant tank structure 280 kg; fuel pumps and boost/transfer units 90 kg; fuel lines, hoses, valves and fittings 110 kg; fuel filters, sensors and quantity measurement 55 kg; fuel isolation/emergency shutoff hardware 75 kg; venting, surge and inerting/fire-protection provisions 70 kg; unusable/residual fuel allocation 120 kg.

Rapid Selective Separation System — 1,000 kg: Primary capsule release mechanisms and locks 240 kg; independent backup release mechanisms 150 kg; local structural reinforcement and release fittings 160 kg; independent initiators/actuators 90 kg; release-state sensors and monitoring 65 kg; independent power/signalling and protected wiring 70 kg; fuel/energy isolation interfaces 55 kg; retention, inhibit and safety-interlock hardware 50 kg; configuration growth allowance 20 kg.

Avionics / Electrical / Hydraulic / Thermal / Control — 1,600 kg: Flight-control computers/control electronics 180 kg; flight-control actuators/servo hardware 210 kg; navigation/air-data/flight sensors 100 kg; communications/mission avionics 90 kg; electrical generation/distribution 190 kg; batteries/emergency electrical interfaces 90 kg; wiring/harnesses/connectors 170 kg; hydraulic pumps/reservoirs/lines/actuators 180 kg; thermal management/cooling 100 kg; environmental control/ventilation 70 kg; fire detection/warning/suppression interfaces 55 kg; health monitoring/data recording 45 kg; lighting/service/misc. 20 kg.

Operating Fluids & Consumables — 320 kg: Engine/transmission oil and other operating fluids 180 kg; hydraulic fluid 35 kg; coolant/thermal working fluid 30 kg; other trapped/operating consumables 55 kg.

23. CRITICAL OPEN ITEMS — REV13

1. Close the 26,000 kg mass model using component-level estimates and uncertainty.
2. Close the 10-12 occupant capsule geometry and bottom-up mass (Document 113).
3. Select/size main rotor blade geometry, RPM, solidity, control system and transmission.
4. Close tail rotor sizing against torque and directional-control requirements.
5. Select candidate engines and close total installed power and OEI performance.
6. Develop 1,000 km mission fuel model including reserves.
7. Develop full 6-DOF aircraft/capsule/released-component model, prioritizing Stage S3 (§13) as the least-precedented item.
8. Define the exact set of jettisonable non-rotor components for each emergency state.
9. Prove safe separation trajectories and post-jettison control for each candidate component.
10. Develop independent automatic and manual command architecture and common-cause analysis.
11. Define Safety Kernel requirements and deterministic implementation boundary.
12. Size the 3-primary/2-reserve canopy system and establish canopy failure tolerance (Document 159), including the low-altitude deployment envelope flagged as [OPEN — CRITICAL] in §14.
13. Evaluate optional residual-airframe parachute versus added mass, drag and deployment risk.
14. Size the eight landing supports and energy absorbers, including asymmetric failures.
15. Perform capsule structural, crash, fatigue and damage-tolerance analyses.
16. Define occupant injury criteria and test instrumentation.
17. Define emergency power and post-separation electrical architecture.
18. Define fire, fuel isolation and thermal protection architecture.
19. Define emergency egress, rescue, locator and communication provisions.
20. Establish certification basis and verification matrix.
21. Commission a professional prior-art search specifically against IP-A, IP-B and IP-C (§17, §19).
22. Conduct independent structural sanity-check of the capsule mass budget (§8.1).
23. Initiate the rotor-separation feasibility research track (Document 080) as a standalone, non-core workstream, should main-rotor separation be pursued in a future revision.

These open items are not weaknesses to be hidden; they are the correct engineering gates that separate a credible conceptual IP baseline from an unsupported performance claim.

APPENDIX A — LICENSING PACKAGE (retained from Rev12, unchanged in substance, cross-references updated)

A.1 Executive Licensing Proposition

HSC Generation 1 is positioned as a licensable conceptual and architectural IP package centred on an occupied survival capsule that can become the final landing boundary in a catastrophic emergency. The Rev13 baseline defines a 26-ton-class single-main-rotor helicopter, a 26.0 m main rotor target, a 4.0 m tail-rotor target, a 1,000 km range target,

controlled non-rotor selective separation, deterministic safety authorization, and multi-mode recovery.

Core IP proposition: the occupied survival boundary is treated as a dedicated survival system rather than only as a conventional cabin. System-level differentiator: survival is organised as a chain of detection, authorization, selective separation, capsule protection, recovery and post-event survival. Commercial proposition: a licensee receives access to an architectural starting point and controlled IP-development framework, not a certified production helicopter. License value should be assessed against the reusable architecture, defined interfaces, the three primary candidate IP claims (§17.1), the disclosed risk register (§18) and verification roadmap, and technology-transfer potential.

Status: [PROPOSED] Additive commercial/IP interpretation. No new performance claim is created by this section.

A.2 Architectural Differentiation Map

Occupied survival capsule as final landing boundary: HSH defines this as a core architectural requirement. State-dependent selective non-rotor separation: HSH explicitly defines release as an authorized emergency mode rather than an indiscriminate jettison command. Deterministic Safety Kernel: the baseline establishes a safety authorization boundary between complex prediction/decision functions and physical survival actuation. Six-layer survival chain: initiation, independent backup/manual authority, selective separation, direct-impact survival and recovery/post-event survival are treated as distinct functional layers. Partial/asymmetric separation: the baseline explicitly treats incomplete release as a first-class emergency state requiring sensing, analysis and degraded recovery logic. Independent post-separation capability: the capsule domain includes independent emergency power, sensing, locator, communications and essential life-support provisions. Multi-mode recovery: controlled flight recovery and autorotation remain available where appropriate, while separation and capsule recovery form additional emergency modes.

Status: [PROPOSED] Additive commercial/IP interpretation.

A.3 IP Architecture / Value-Surface Map

See §17 for the Rev13-narrowed claim set (IP-A, IP-B, IP-C) and §19 for the corresponding prior-art posture. Legal note: novelty, patentability, claim scope, validity and freedom-to-operate are not established by the baseline or this package. Professional prior-art and claim analysis remain required before filing.

A.4 Licensee Due-Diligence & Maturity Roadmap

A serious licensee should be able to distinguish what is defined from what is still open. Architecture: defined at conceptual baseline level, with the Rev13 correction of the main-rotor dependency. Requirements: top-level requirements and targets are defined, with allocation and closure still required. Mass: 26,000 kg aircraft accounting and 7,000 kg capsule allocation are planning values requiring bottom-up closure (§8.1, §23). Dynamics: 6-DOF emergency dynamics, separation trajectories, partial/asymmetric release and parachute inflation remain open verification work, with Stage S3 (§13) as the priority gap. Separation: selective and staged architecture is defined conceptually and now explicitly excludes the main rotor; release mechanism reliability and safe trajectories require analysis and testing. Recovery: the 3+2 canopy architecture (§14) is a preliminary study concept requiring sizing, deployment and failure-case testing, with the low-altitude envelope as a flagged critical open item. Structures: capsule, attachments, landing supports and occupant protection require analytical and physical substantiation. Certification: the applicable certification basis and formal qualification evidence are not yet established. IP: three candidate surfaces are identified and prior-art-aware (§17, §19); novelty and freedom-to-operate require professional legal analysis. Gate logic: G0 through G8 as defined in §16, with G0 the only gate presently closed.

Status: [PROPOSED] Additive commercial/IP interpretation.

A.5 Staged Separation Architecture (commercial summary)

For commercial readability, the corrected separation architecture (§13, §12) can be expressed as three functional stages: **Stage A — Hazard Reduction** (isolate and separate designated non-rotor high-mass, rotating or fire-hazard components while maintaining controlled capsule load paths); **Stage B — Capsule Liberation** (release remaining designated airframe-to-capsule attachments so the occupied capsule can become the independent final landing boundary); **Stage C — Clearance Verification & Recovery** (positively determine capsule-clear status, identify residual attached components, and authorize the recovery system only within its validated envelope). The per-stage timing given in §13 remains a TARGET/OPEN development objective, not demonstrated performance. Partial and asymmetric release must remain explicitly represented in the state model at every stage.

Status: [PROPOSED] Additive commercial/IP interpretation.

A.6 Technology Platform & Future Product Family

The underlying survival architecture can be treated as a potential technology platform, while keeping future variants clearly separated from the Generation 1 engineering baseline: HSH-G1 Heavy (the present baseline); HSH Utility; HSH Rescue; HSH Expedition; HSH Special Mission. These future families are conceptual opportunities only, not implied to be engineered, tested, certified or commercially available. The reusable platform opportunity lies in architecture, interfaces, safety logic, survival-state management and verification methods, while vehicle-specific hardware must be re-engineered for each derivative.

Status: [PROPOSED] Additive commercial/IP interpretation.

A.7 Strategic Licensing Path

Phase 1 — NDA: establish confidentiality and permitted evaluation scope. Phase 2 — Controlled Technical Data Room: provide the Rev13 baseline, this licensing package and the controlled document-tree material appropriate to the evaluation stage. Phase 3 — Technical/IP Due Diligence: review requirements, mass model, interfaces, separation concept, safety architecture, the Rev13 Risk Register (§18), verification roadmap and the three primary candidate IP surfaces (§17.1). Phase 4 — Exclusive Evaluation Option: if commercially appropriate, define a time-limited evaluation or option period with controlled technical access. Phase 5 — License Term Sheet: define territory, field of use, exclusivity, milestones, economics, sublicensing and improvement rights. Phase 6 — Engineering Development Agreement: allocate responsibility for simulation, testing, qualification, certification and product definition. Phase 7 — Commercial License/Product Development: transition the controlled IP package into the licensee's engineering and industrialisation system. Commercial terms remain contractual matters and are not established by this engineering document.

Status: [PROPOSED] Additive commercial/IP interpretation.

A.8 Licensee's First 10 Questions

1. **What exactly is the IP?** Three primary candidate surfaces (§17.1), explicitly narrowed from Rev12's nine, with supporting system architecture.
2. **What is actually defined today?** The architecture, major interfaces, safety philosophy, emergency state machine, corrected non-rotor separation logic and verification structure, defined at conceptual-baseline level.
3. **What is not yet proven?** Mass closure, detailed dynamics (especially Stage S3), separation reliability, parachute performance at low altitude, structural qualification, flight behaviour, certification and patentability — all disclosed explicitly in §18 and §23.
4. **What is the core differentiator?** The occupied survival capsule combined with controlled, non-rotor, state-dependent separation and deterministic safety authorization.
5. **Can the architecture scale?** The survival principles can be considered a platform opportunity, but every derivative requires independent engineering closure and certification.
6. **What prevents uncontrolled separation?** Mechanical retention, electrical/logic inhibits, independent command paths, defined arming, release-state sensing and safety preconditions (§6, §7.3).
7. **What happens if separation is incomplete?** Partial/asymmetric release is explicitly treated as an emergency state at every stage, requiring state sensing and a defined degraded recovery strategy.
8. **How does AI interact with safety?** AI may support detection, prediction and candidate-action ranking, while deterministic safety logic (§6) remains the sole authorization boundary — this is IP-A, the strongest claim in the package.
9. **How does the buyer know when the concept becomes credible?** Evidence gates G0–G8 (§16) define the progression; Rev13 states plainly that only G0 is presently closed.
10. **What happens after licensing?** The proposed path (§A.7) moves from controlled data-room evaluation into a term sheet and then a structured engineering-development programme.

Status: [PROPOSED] Additive commercial/IP interpretation.

END OF DOCUMENT 00 — HSC GENERATION 1 | REVISION 13 Capsule: 7,000 kg | Vehicle: 26 t | Main rotor: 26.0 m | Tail rotor: 4.0 m Founder: Amirbahador Atai | Not a certification approval